



Predicting the Length of Hospital Stays

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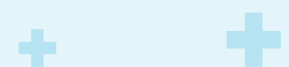


Project Overview: Hospital Stay Prediction

This case study will focus on duration of hospital stay data, and on creating predictive models on the duration of patients length of stay in days.

Main Objectives:

- **Objective 1:** Build an interpretable multiple linear regression (MLR) model to identify key predictors of patient length of stay.
- **Objective 2:** Improve predictions using complex MLR with interactions/polynomials and a nonparametric model (kNN).
- **Overall:** Our aim is to help hospitals reduce unnecessary patient days by finding the most influential predictors of patient length of stay.



Why this matters

- Hospital stays are expensive and time-sensitive.
- Predicting length of stay helps improve staffing, cost control, and patient outcomes.
- AI and statistical models support better forecasting and decision-making.



Data Description Table

Data: 113 Hospitals, 11 predictors

Variable Name	Data Type	Description
Length of Stay	numeric	Average length of stay (in days)
Age	numeric	Average age of patients (years)
Infection risk	numeric	Average estimated probability of hospital infection
Routine culturing ratio	numeric	Ratio of # of cultures performed to number of patients without symptoms of infection times 100
Routine chest X-ray ratio	numeric	Ratio of # of cultures performed to number of patients without symptoms of pneumonia time 100
Available Facilities	numeric	Percent of 35 potential facilities and services that are provided by the hospital
Average Patients	integer	Average number of patients in hospital per day
Average Nurses	integer	Average number of full time nurses
Number of beds	integer	Average number of beds
Medical School Affiliation	categorical	Medical school affiliation (1=Yes 2=No)
Region	categorical	Geographic region: (1=NE 2=NC 3=S 4=W)

80/20 TRAIN-TEST SPLIT

80%
TRAIN

20%
TEST



Exploratory Data Analysis

Summary Statistics

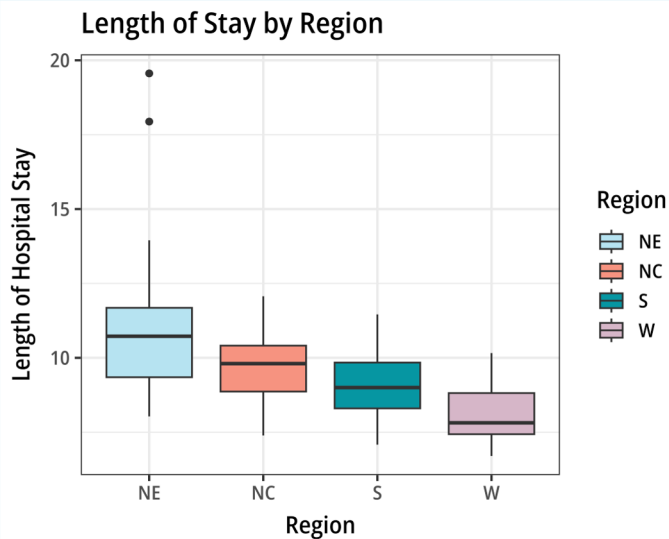
Data: 113 Hospitals, 9 numeric/integer predictors

Variable	Length of Stay	Age	Infection Risk	Routine culturing ratio	Routine chest X-ray ratio	Number of Beds	Average # of Patients	Average # Full-Time Nurses	Percent SerFac
Min	6.7	38.8	1.3	1.6	39.6	29	20	14	5.7
1st Qu.	8.34	50.9	3.7	8.4	69.5	106	68	66	31.4
Median	9.42	53.2	4.4	14.1	82.3	186	143	132	42.9
Mean	9.648	53.23	4.355	15.79	81.63	252.2	191.4	173.2	43.16
3rd Qu.	10.47	56.2	5.2	20.3	94.1	312	252	218	54.3
Max	19.56	65.9	7.8	60.5	133.5	835	791	656	80

Categorical Variables

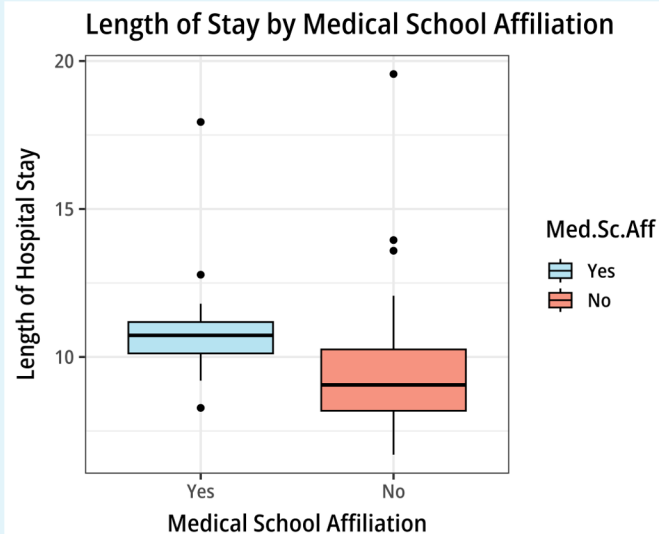
Region

	NE	NC	S	W
Count	28	32	37	16
Percent	25%	28%	33%	14%



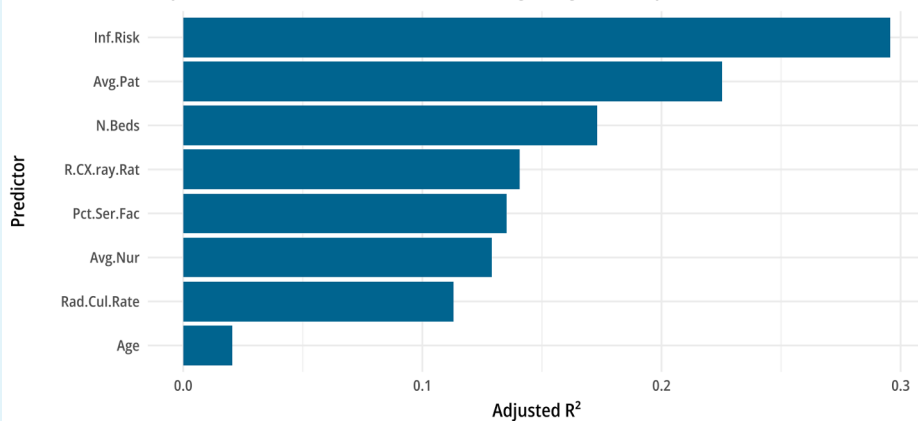
Medical School Affiliation

	Yes	No
Count	17	96
Percent	15%	85%

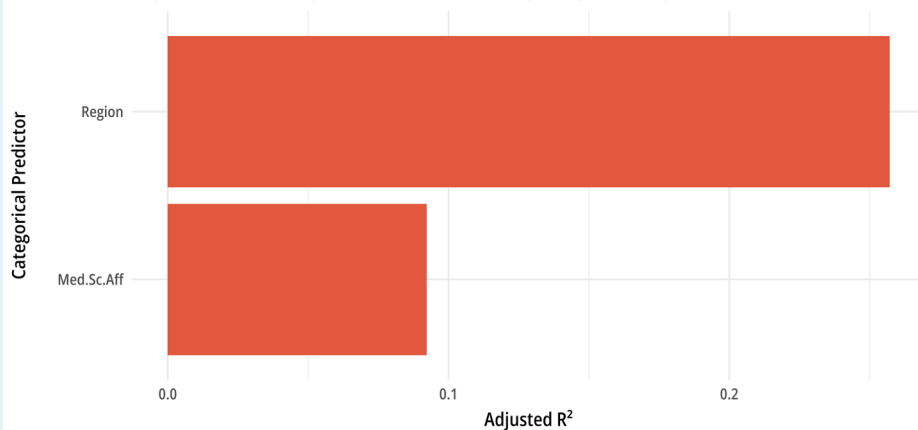


EDA Continued

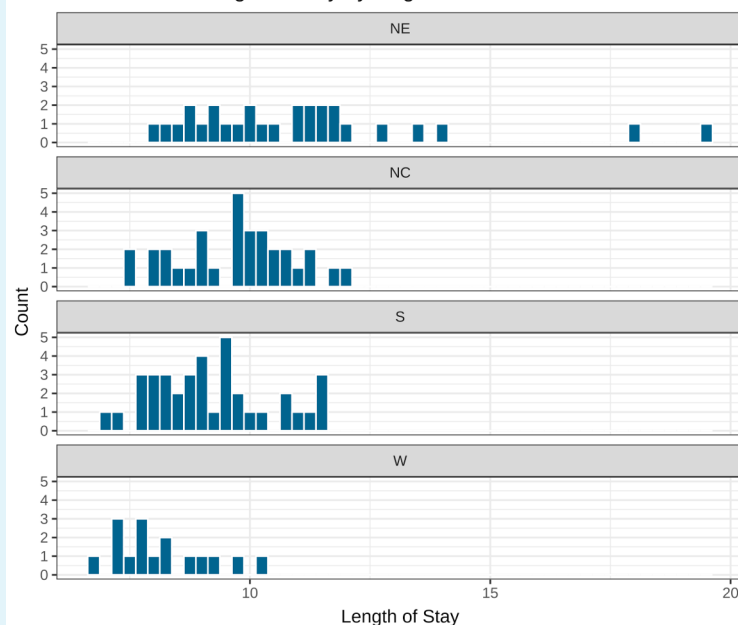
Adjusted R^2 of Numeric Predictors for $\log(\text{Length of Stay})$



Adjusted R^2 of Categorical Predictors for $\log(\text{Length of Stay})$

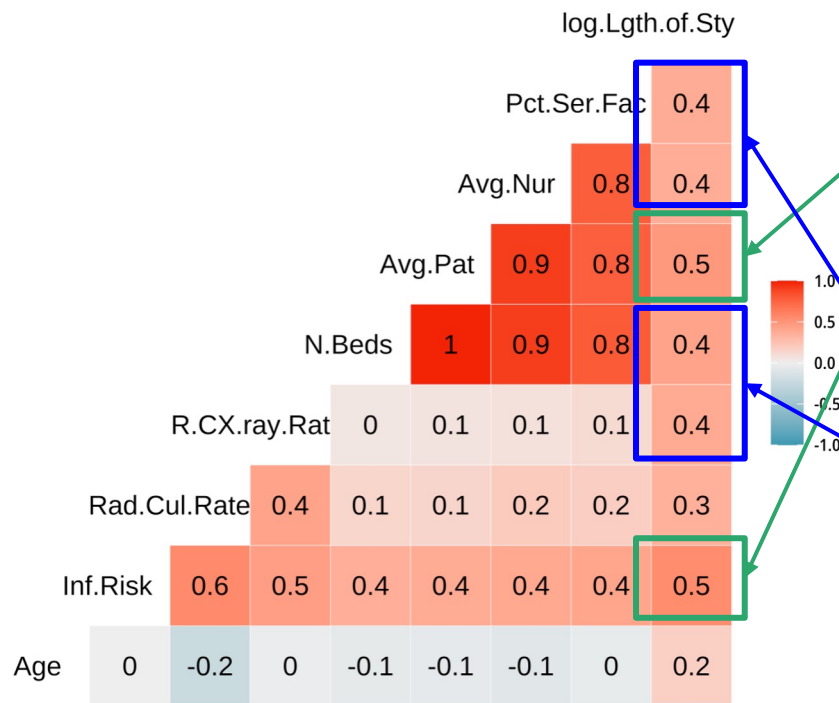


Distribution of Length of Stay by Region



Correlation Matrix

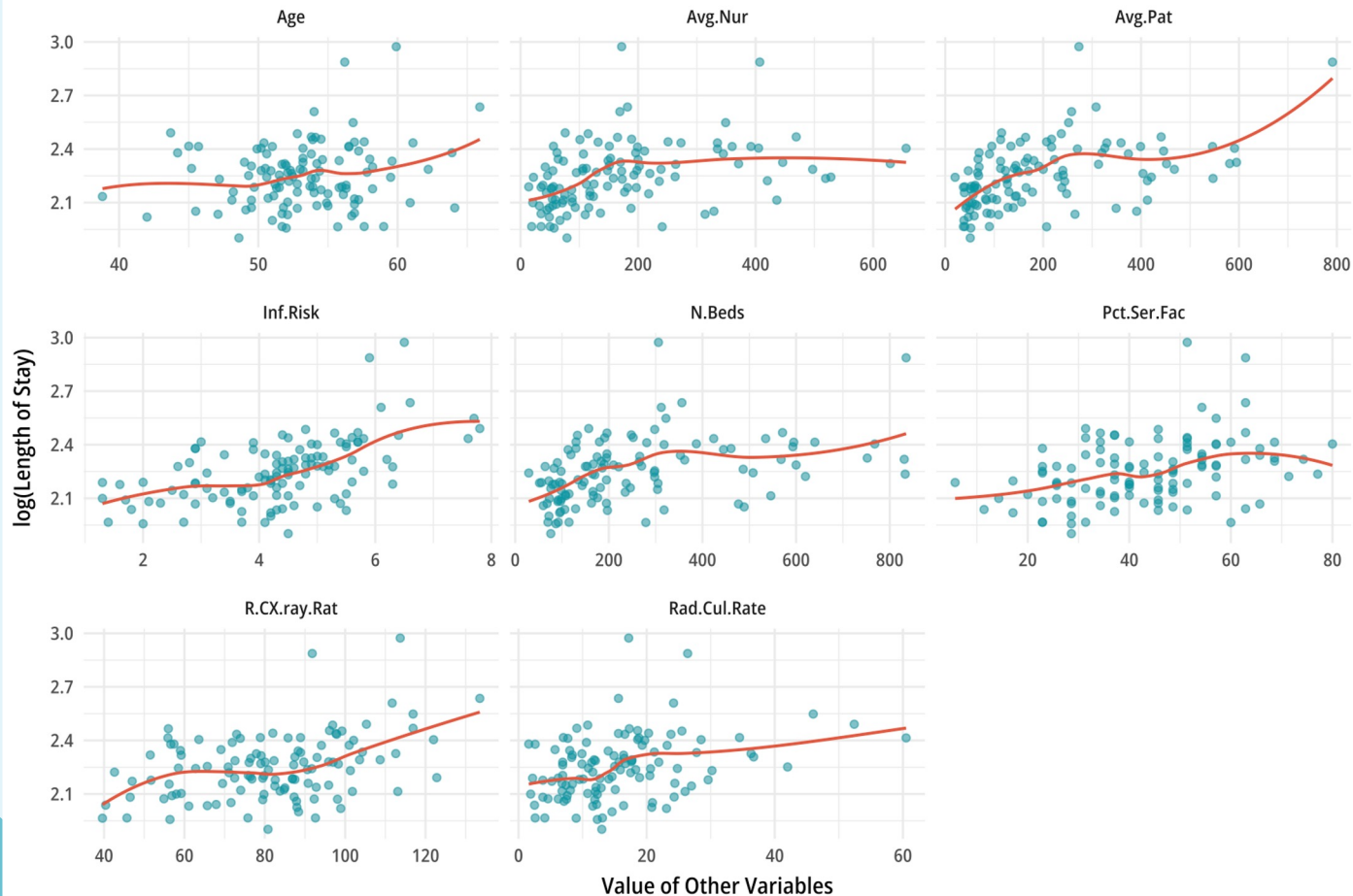
Correlation Matrix - log(Length of Stay)



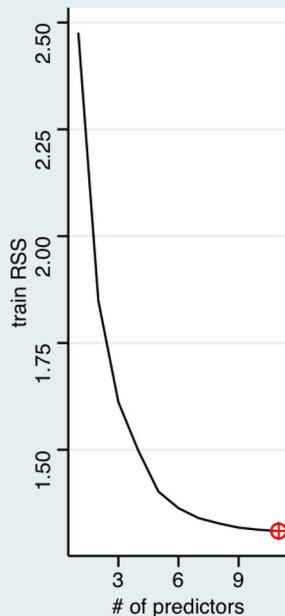
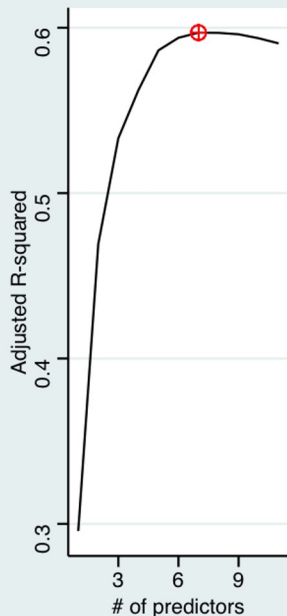
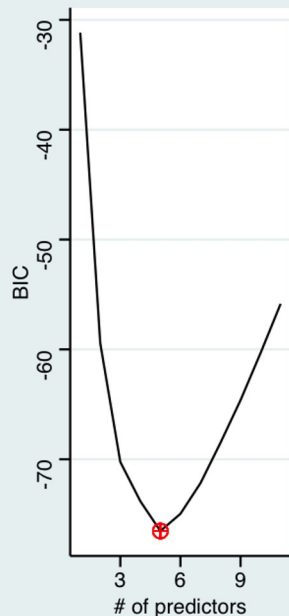
Initial top predictors of length of stay to consider

Initial secondary predictors of length of stay to consider

log(Length of Stay) vs. Other Hospital Variables



Forward Stepwise Variable Selection



Variables
log(Lgth.of.Sty)
Age
Inf.Risk
Region
Avg.Pat
Avg.Nur

BIC (Bayesian Information Criterion): we can see our best model fit is **5 predictors**.

Adjusted R-squared: as we increase our number of predictors up to seven, our R-squared rises. After 7 we see our score degrade as the model overfits

RSS(Residual Sum of Squares): We see a constant decline up until our minimum at 11 predictors.

Simple MLR Model (Additive Model)

Variables Included:

- Response: $\log(\text{Lgth.of.Sty})$
- Age
- Infection Risk
- Region
- Average # of patients in hospital per day
- Average # of full-time nurses

MODEL



Simple MLR

VARIABLE SELECTION

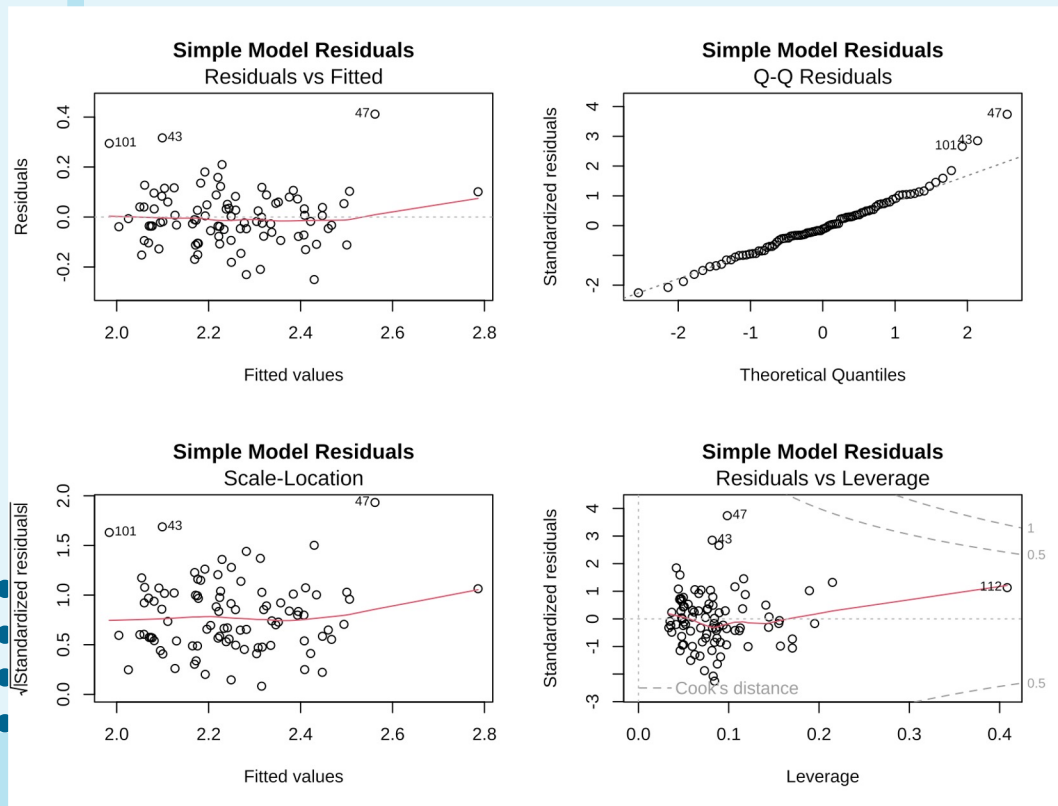
To identify the variables used in our first model, we used Forward and Stepwise Variable Selection: we evaluated three different plots when identifying our selection. **BIC, Adj.R², and the RSS plot**

PERFORMANCE

- **R-squared:** 0.5977
- **Adj R-squared:** 0.5645
- **RSE:** 0.1171
- **F-Stat** 18.04
- **P-value:** 1.76e-14



Simple MLR: Residual Diagnostics



Residuals vs Fitted: Residuals appear randomly scattered with no clear trend suggesting linearity and constant variance are mostly reasonable. The residuals are somewhat more gathered on the left side, so we will proceed with caution.

Normal Q-Q: Residuals mostly have a linear trend with a slight tail deviation on the right side, therefore the normality assumption is mostly satisfied.

Scale-Location: Spread of residuals is fairly even — no strong evidence of heteroscedasticity.

Residuals vs Leverage: Most points have low leverage; one high-leverage point may warrant attention but is not outside the Cook's D cut-off of 2. We will move forward with the MLR model.

Additive MLR

Equation

$\text{lm}(\text{formula} = \log(\text{Lgth.of.Sty}) \sim \text{Region} + \text{Avg.Pat} + \text{Age} + \text{Inf.Risk} + \text{Avg.Nur}, \text{data} = \text{train})$

Interpretation of Infection Risk

For every one unit increase in Infection Risk, there is an estimated multiplicative change of $e^{0.0526} = 1.054$ in the median hospital Length of Stay. That is, each 1-unit increase in infection risk is associated with a 5.4% increase in the median length of stay, holding all other variables constant.

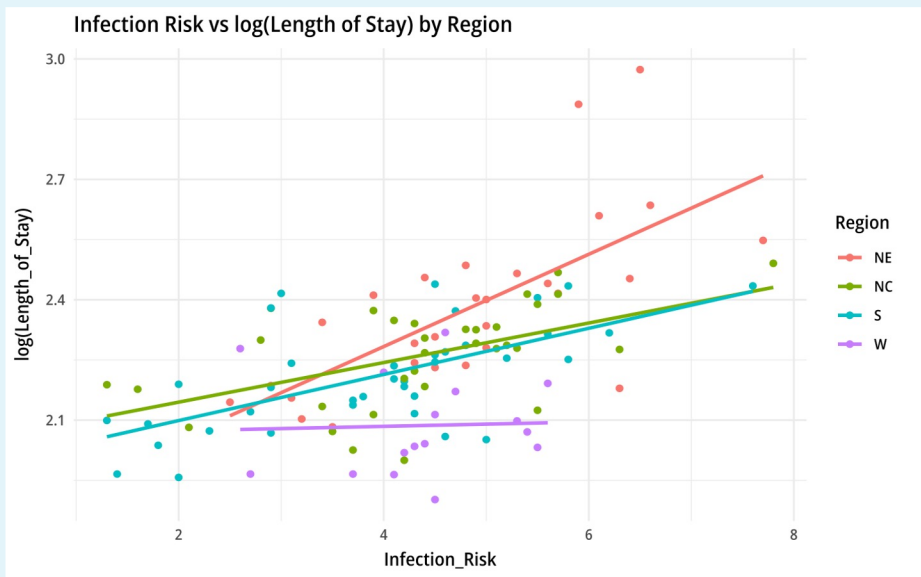
Infection Risk 95% Confidence Interval

Based on the paired bootstrap, we are 95% confident that a 1-unit increase in Infection Risk is associated with a ($e^{0.0328} = 1.033$, $e^{0.0729} = 1.076$) 3.3% to 7.6% increase in the median length of hospital stay, holding all other variables constant.

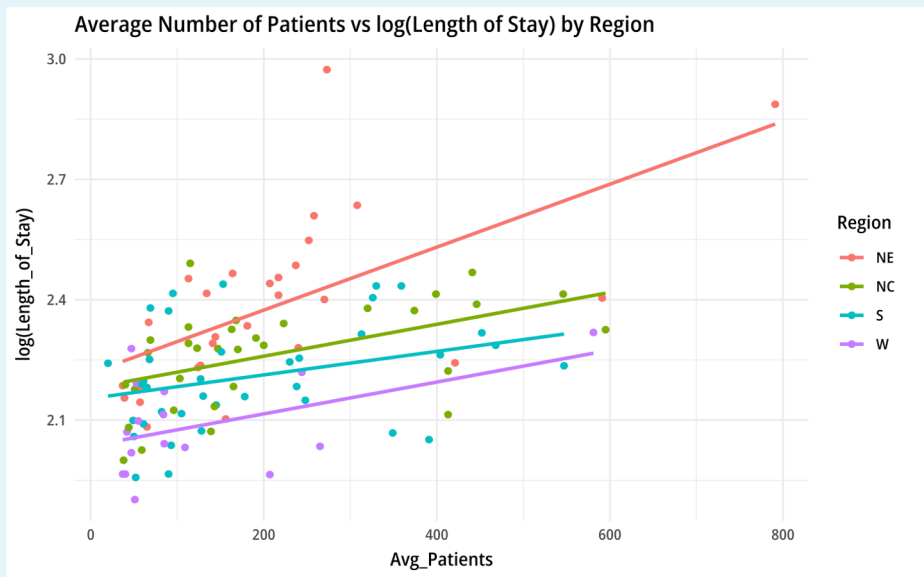


Interactions

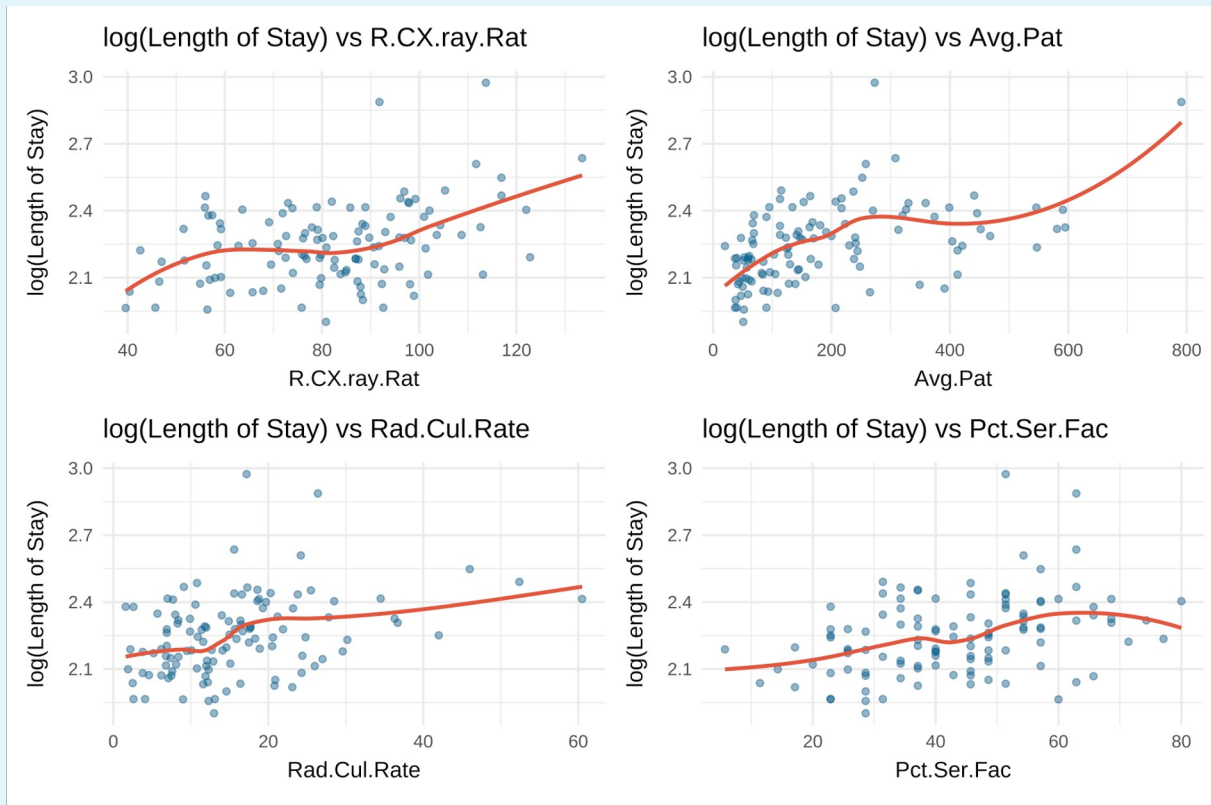
Infection Risk and Region



Average Number of Patients and Region



Polynomial Terms Added



Complex MLR Model

Variables Included:

Main Effects & Polynomial Terms

- Age²
- Avg.Pat³, Avg.Pat⁴
- Rad.Cul.Rate⁵
- Pct.Ser.Fac, Pct.Ser.Fac³
- R.CX.ray.Rat, R.CX.ray.Rat^{2, 3, 4}

Interaction Terms

- Inf.Risk × Region
- Avg.Pat × Region
- Avg.Nur × Region

MODEL



Complex MLR

VARIABLE SELECTION

To help identify the variables used in our second model, we used a Lasso Regression model:

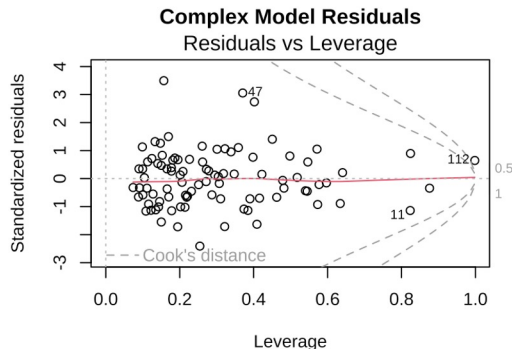
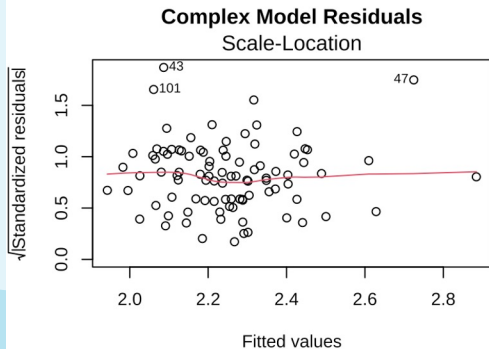
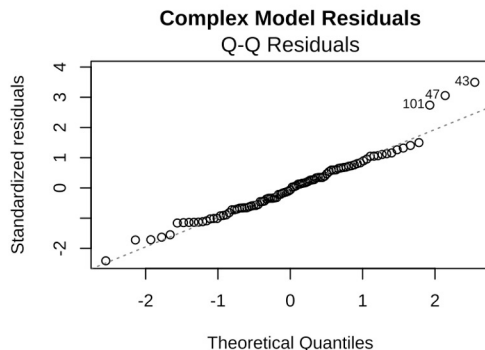
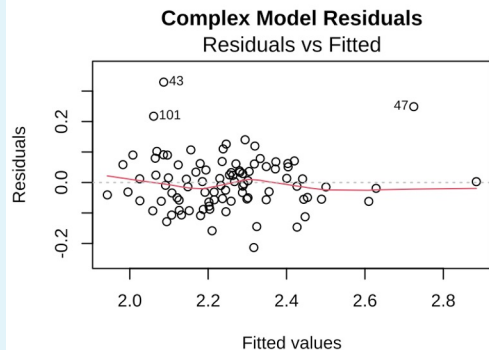
This allowed us to take another approach to testing the performance of our final model

PERFORMANCE

- R-squared: 0.7758
- Adj R-squared: 0.6921
- RSE: 0.1014
- F-Stat 9.273
- P-value: 1.824e-13



Residual Diagnostics



Residuals vs Fitted: Residuals are somewhat bunched in the middle left, but somewhat randomly scattered with no clear trend. Linearity and constant variance are reasonable, but we must proceed with caution.

Normal Q-Q: Residuals mostly have a linear trend with some deviation on the right tail end, therefore the normality assumption is mostly satisfied.

Scale-Location: Spread of residuals is fairly even — no strong evidence of heteroscedasticity.

Residuals vs Leverage: Most points have low leverage; one high-leverage point may warrant attention as it is right on the Cook's D cut-off of 2. We will move forward with the MLR model.

k-NN Model

Variables Included:

- Response: $\log(\text{Lgth.of.Sty})$
- Age
- Infection Risk
- Region
- Average # of patients in hospital per day
- Average # of full-time nurses
- Routine chest X-ray ratio

MODEL



K-NN model

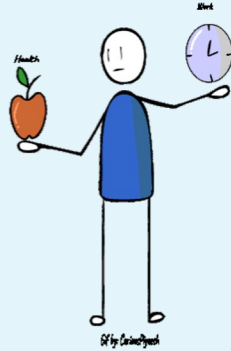
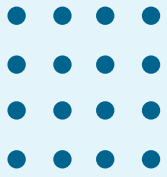
VARIABLE SELECTION

We used k-fold cross-validation with 10 folds and 10 repeats and found the best k-value to be **k=5**

PERFORMANCE

RMSE: 0.1275
R Squared: 0.352
MSE: 0.016





Model Comparison



Model Metric Comparison

	Simple MLR 	Complex MLR 	k-NN 	Best Model 
MSE	0.012	0.009	0.016	Complex MLR
RMSE	0.1075	0.0996	0.1275	Complex MLR
Adjusted R2	0.597	0.669	0.352	Complex MLR
MAE	0.085	0.077	0.1040	Complex MLR
AIC	-127.142	-138.314	N/A	Complex MLR
BIC	-104.349	-69.934	N/A	Simple MLR

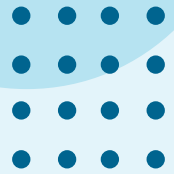


General Findings and Final Takeaways

- **Infection Risk** was a significant predictor of hospital length of stay across all models.
- The **Simple MLR** provided a good baseline with clear interpretability and meaningful inference.
- The **Complex MLR** (with interactions and polynomial terms) improved predictive performance and metrics substantially.
- The **k-NN model**, while nonparametric and less interpretable, resulted in worse error metrics compared to our regression models. While it offers flexibility, its predictive performance was not as strong and lacks the interpretability of the MLR models.



Conclusions



Scope of Inference

- This is an observational study, so we can only identify associations, not causal relationships.
 - Additionally, since the 113 hospitals may not represent a random or nationally representative sample, our inferences should be limited to these specific 113 hospitals and not generalized beyond them.
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If given more time:

- Explore additional nonparametric methods (e.g., random forest, regression tree) for robustness.
 - Investigate potential multicollinearity solutions or dimension reduction (e.g., VIF, 1:1 relationship between Avg. Patients and Number of Beds).
 - Use external validation on a second dataset to further confirm model generalizability.
 - Study how temporal trends (e.g., before/after COVID, seasonal trends) may influence current hospital practices.
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For the future....

We recommend:

The Complex/Nonadditive MLR

Thanks!

Do you have any questions?



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